Unit 3.01: Gradient Descent & Optimization

Q1. In gradient descent, a weight currently has value w = 10 . If the gradient of the loss is frac{partial L}{partial w} = 4 and the learning rate is eta = 0.1 , find the updated weight w' .

[ ] A: 10.40
[x] B: 9.60
[ ] C: 6.00
[ ] D: 9.90

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 10 - 0.14 = 10 - 0.40 = 9.60 .

Q2. In gradient descent, a weight currently has value w = 20 . If the gradient of the loss is frac{partial L}{partial w} = 8 and the learning rate is eta = 0.2 , find the updated weight w' .

[ ] A: 12.00
[ ] B: 21.60
[x] C: 18.40
[ ] D: 19.80

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 20 - 0.28 = 20 - 1.60 = 18.40 .

Q3. In gradient descent, a weight currently has value w = 30 . If the gradient of the loss is frac{partial L}{partial w} = 12 and the learning rate is eta = 0.3 , find the updated weight w' .

[ ] A: 18.00
[ ] B: 33.60
[x] C: 26.40
[ ] D: 29.70

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 30 - 0.312 = 30 - 3.60 = 26.40 .

Q4. In gradient descent, a weight currently has value w = 40 . If the gradient of the loss is frac{partial L}{partial w} = 16 and the learning rate is eta = 0.4 , find the updated weight w' .

[ ] A: 24.00
[ ] B: 46.40
[x] C: 33.60
[ ] D: 39.60

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 40 - 0.416 = 40 - 6.40 = 33.60 .

Q5. In gradient descent, a weight currently has value w = 50 . If the gradient of the loss is frac{partial L}{partial w} = 20 and the learning rate is eta = 0.5 , find the updated weight w' .

[ ] A: 30.00
[ ] B: 60.00
[x] C: 40.00
[ ] D: 49.50

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 50 - 0.520 = 50 - 10.00 = 40.00 .

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Q6. In gradient descent, a weight currently has value w = 60 . If the gradient of the loss is frac{partial L}{partial w} = 24 and the learning rate is eta = 0.6 , find the updated weight w' .

[ ] A: 36.00
[ ] B: 74.40
[x] C: 45.60
[ ] D: 59.40

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 60 - 0.624 = 60 - 14.40 = 45.60 .

Q7. Consider the scalar loss function Lw = 2w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 2 .

[ ] A: nabla L2 = 16
[ ] B: nabla L2 = 8
[x] C: nabla L2 = 0
[ ] D: nabla L2 = -4

Solution: Differentiate with respect to w : frac{dL}{dw} = 22w - 8 = 4w - 8 . At w = 2 : nabla L = 42 - 8 = 8 - 8 = 0 .

Q8. Consider the scalar loss function Lw = 3w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 4 .

[x] A: nabla L4 = 16
[ ] B: nabla L4 = 24
[ ] C: nabla L4 = 32
[ ] D: nabla L4 = 4

Solution: Differentiate with respect to w : frac{dL}{dw} = 23w - 8 = 6w - 8 . At w = 4 : nabla L = 64 - 8 = 24 - 8 = 16 .